

Table 2. [All-in-one] Complete comparison of restoration quality, source-identity recognition, and lightweight efficiency on the PLAMD test set. The identity gallery contains 256 clean sources. Red and blue indicate the best and second-best values among the restored methods; lower is better for LPIPS, EER, Params, MACs, latency, and peak VRAM.

Method / cohort		Restoration quality			Identity recognition					Lightweight / efficiency					
Method	Cohort	PSNR ↑	SSIM ↑	LPIPS ↓	Top-1 ↑	Top-5 ↑	SameCat	Top-1 ↑	EER ↓	AUC ↑	Params (M) ↓	MACs (G) ↓	Latency (ms) ↓	VRAM (GB) ↓	Throughput ↑
Degraded input	all	—	—	—	0.3030	0.5106	0.3622	0.1687	0.9077		—	—	—	—	—
A5 (Ours)	ablation_core	28.7905	0.9458	0.1433	0.6981	0.8605	0.7344	0.0799	0.9749		2.63	25.52	46.42	0.20	21.54
	dehazeformer	28.3717	0.9433	0.1376	0.6708	0.8354	0.7193	0.0942	0.9692		25.45	275.15	79.12	0.37	12.64
	restormer	28.1046	0.9433	0.1496	0.6562	0.8382	0.7098	0.0902	0.9702		26.13	140.99	84.66	0.66	11.81
	promptir	27.5255	0.9381	0.1579	0.6334	0.8276	0.6858	0.0944	0.9674		35.59	158.14	91.64	0.70	10.91
	airnet	24.7851	0.8793	0.2064	0.4442	0.6708	0.5017	0.1357	0.9395		8.93	301.27	167.45	0.55	5.97
	clearair	21.0695	0.8360	0.2876	0.2584	0.4860	0.3131	0.1896	0.8917		32.21	134.99	81.17	0.68	12.32
	dfpir	27.2762	0.9323	0.1717	0.5954	0.7896	0.6445	0.0998	0.9625		26.13	140.99	84.67	0.66	11.81
	ffanet	26.6121	0.9206	0.2058	0.5301	0.7718	0.5993	0.1082	0.9557		1.06	68.38	12.43	0.12	80.44
	grl	20.4327	0.7615	0.3502	0.3092	0.5134	0.3605	0.1678	0.9091		5.11	316.39	838.44	0.60	1.19
	prenet	24.3834	0.8733	0.2360	0.3711	0.5938	0.4369	0.1660	0.9132		0.46	30.31	4.32	0.30	231.36
	r2r	26.1536	0.9176	0.2359	0.5519	0.7589	0.6071	0.1117	0.9554		30.92	16.65	28.83	0.19	34.69
	swinir	24.7790	0.8850	0.2310	0.3912	0.6138	0.4453	0.1591	0.9194		3.14	204.53	145.99	0.86	6.85
	transweather	23.7907	0.8781	0.2745	0.3181	0.5809	0.3956	0.1522	0.9254		38.05	6.13	9.93	0.18	100.72
	uformer	25.6141	0.9097	0.2153	0.4883	0.6775	0.5469	0.1383	0.9366		20.63	41.09	38.44	0.57	26.02

Identity metrics are from the official frozen-verifier aggregate on the same 256-source clean gallery. Efficiency metrics are measured at the project benchmark input size; A5 uses the corresponding dim-48 anchored architecture record from `efficiency_records.csv`. The two standalone dim-variant rows are intentionally omitted as requested; their A5-equivalent efficiency record is not duplicated as a separate method.